Tribhuvan University

Faculty of Humanities and Social Sciences

ONLINE SHOE STORE

A PROJECT REPORT

Submitted to

Department of Computer Application

Birendra Multiple Campus

Bharatpur-10, Nepal

In partial fulfillment of the requirements for the Bachelor in Computer Application

Submitted by

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BCA 6th Semester

Under the Supervision of

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **SAGAR BHATTARAI** and **DIKSHYANT KHATRI CHHETRI** entitled **"ONLINE SHOE STORE"** in partial fulfillment of the requirement for the degree of Bachelor in Computer Application is recommended for the final evaluation.

.....

Mr. Narayan Prasad Dahal

PROJECT SUPERVISOR

Birendra Multiple Campus

Bharatpur, Nepal

Tribhuvan University

Faculty of Humanities and Social Sciences

Birendra Multiple Campus

LETTER OF APPROVAL

This is to certify that this project prepared by **SAGAR BHATTARAI** AND **DIKSHYANT KHATRI CHHETRI** entitled "**ONLINE SHOE STORE"** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

_____ **Supervisor** Narayan Prasad Dahal BCA, Birendra Multiple Campus Bharatpur, Nepal	_____ **Program Director** Soba Raj Poudel BCA, Birendra Multiple Campus Bharatpur, Nepal
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ABSTRACT

The Online shoe store is a web-based e-commerce application developed to provide a convenient and user-friendly platform for online shoe shopping. The system allows customers to browse shoes based on categories such as male, female, and kids, view product details, and place orders securely. It supports user registration and authentication to ensure personalized access, order tracking, and purchase history management.

The system includes two major modules: User Module and Admin Module. The User Module enables customers to explore products, add items to the cart, and complete purchases using an integrated digital payment system (eSewa) or cash on delivery payment methods. The Admin Module provides administrators with full control over product management, order processing, user management, and payment status monitoring.

The application is developed using PHP and MySQL for backend operations, with HTML, CSS, and JavaScript for frontend design to ensure responsiveness and usability. Secure payment handling, session management, and role-based authentication enhance the system's reliability and security. The Shoe Store aims to reduce manual work, improve business efficiency, and offer a seamless online shopping experience for customers.

Keywords:

Online shoe store, place order, order tracking, purchase history management, esewa, payment handling, session management.

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Yours sincerely,

Sagar Bhattarai

Dikshyant Khatri Chherti

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LIST OF ABBREVIATIONS

Abbreviation	Definition
CRUD	Create, Read, Update and Delete
CSS	Cascading Style Sheet
DFD	Data Flow Diagram
ER	Entity Relationship
HTML	Hypertext Markup Language
JS	Java Script
MySQL	Microsoft Server Structured Query Language
OSS	Online shoe store
XAAMP	Cross-Platform (X), Apache (A), MariaDB (M), PHP (P), and Perl (P).

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CHAPTER 1:
INTRODUCTION

1.1 Introduction

The Online Shoe Store is a web-based e-commerce system that allows customers to buy shoes anytime and from anywhere. Footwear is an important part of daily life, and people need different types of shoes for daily use, sports, work, and other activities. Nowadays, online shopping is very popular because it saves time and provides convenience.

The Online Shoe Store offers a wide variety of shoes for male, female, and children. Customers can easily browse products, view details such as price, size, and stock, and choose their desired shoes. To purchase products, customers must register and log in to the system. After logging in, they can add products to the shopping cart, remove items if needed, and place orders using available payment options.

This system is designed to be easy to use for both beginners and advanced users. The project includes two main modules: Admin and Customer. The Admin can add, update, view, and delete products, as well as manage customer orders. Customers can register, log in, browse shoes, manage their cart, and place orders.

Overall, the Online Shoe Store makes shoe shopping simple, fast, and convenient by allowing customers to purchase shoes from the comfort of their home.

1.2 Problem Statement

Traditional shoe shopping is time-consuming and inconvenient due to limited store hours, travel requirements, and limited product availability. Customers often do not get enough choices in design, size, and stock. At the same time, manual management of products and orders creates difficulties for store owners, leading to errors and poor record keeping. The problem statement highlights some key issues associated with current shoes store that make users uncomfortable. While these systems are advanced, they have certain drawbacks that must be addressed.

1. Traditional shoe shopping is time-consuming and inconvenient.
2. Limited product choices in physical stores.
3. Manual product and order management causes errors.

1.3 Objectives

The main objective of our project is to create a user-friendly and efficient instant shopping tool that can handle multiple users simultaneously. Our goal is to address the issues highlighted in the previous chapter.

1. To save time and easy to purchase.
2. To make easy comparison.
3. To make automate Management

1.4 Scope and Limitation

1.4.1 Scope

The scope of the Online Shoe Store includes developing a web-based system that allows customers to browse shoes, view product details, register, log in, add products to a cart, and place orders online. The project also covers an admin module where the administrator can manage shoe products, categories, prices, stock, and customer orders. The system focuses on providing a simple, secure, and user-friendly interface for online shoe shopping and efficient store management.

1.4.2 Limitation

Some of the limitations of the system are listed below:

- Requires internet access to use the system.
- Online payment depends on third-party payment gateways.
- Delivery time depends on courier services.

1.5 Report Organization

1.5.1 Introduction

This chapter describes the introduction of the built system. It includes five subtopics i.e. Introduction, Problem of statement, Objective, limitations. It describes the early stages of a software development.

1.5.2 Background Study and Literature Review

This chapter describes the Background study and Overview of related existing systems and their pros and cons. It includes 2 topics, Background Study and literature reviews. This chapter focuses on the background of the project as well as the reference taken to start this project.

1.5.3 System Analysis and Design

This chapter focuses on the System Analysis and Design including Requirement Analysis and Feasibility Analysis. It’s analyzed the different requirement of the system, which

describes about the functional, non-functional, feasibility analysis, Entity Relational diagram, Data Flow Diagram, design of the system with system architecture, database schema, and interface design

1.5.4 Implementation and Testing

This Chapter presents the Implementation, Testing and debugging. Implementation consists of tools used and implementation details of modules. Testing consists of test case with unit testing i.e. testing the login form, and register form. It also consists of system testing which is the testing of the entire system.

1.5.5 Conclusion and Future Recommendation

In this Chapter, Conclusion, Lesson learnt and Future Enhancements are briefly explained. In Conclusion we deduce the entire purpose of the project as well as its objectives. In lesson learnt we express the point we have learnt while making this project. In Future Enhancements the unavailable features soon to be added are listed.

CHAPTER 2:
BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Online shopping, or e-commerce, is the process of buying goods or services over the internet. It allows customers to browse, select, and purchase products conveniently from home or any location with internet access. The online footwear market has grown rapidly as more people prefer the convenience of shopping online. Customers can create an account by registering on the website. After registration, they can log in, explore different shoe categories, and select the products they want. Customers can add items to the shopping cart, make changes by adding or removing products, and then complete the purchase through the available payment options.

2.2 Literature Review

E-commerce has transformed how customers shop for footwear by offering convenience and accessibility. One of the main factors influencing adoption is user acceptance and trust. The Technology Acceptance Model suggests that ease of use and usefulness strongly determine consumers' willingness to shop online [1].

Another key concern is secure payments, as customers often hesitate due to fears of fraud or data theft. Research shows that The benefits of secure payment gateways for ecommerce security extend well beyond fraud prevention—they build customer trust, reduce cart abandonment, and ensure that sensitive financial information stays confidential and protected [2].

Many existing e-commerce systems, including well-established platforms like Macy's, Ekjor, and Aadima Store, have significantly contributed to the growth of online retail by providing users with access to a wide range of products and streamlined purchasing processes. However, research and user feedback indicate that these systems still exhibit several limitations that affect overall user satisfaction and operational efficiency. For instance, slow navigation and page loading times have been reported as persistent issues, particularly on mobile devices and during peak traffic periods, which can lead to increased bounce rates and abandoned shopping carts [3]. Additionally, limited stock visibility where users are unable to see accurate information about product availability in real time creates uncertainty and reduces trust in the platform's ability to fulfill orders promptly [4]. Another common shortcoming is the lack of effective personalized recommendations; many systems still rely on basic filtering and search mechanisms rather than leveraging advanced user

behavior analytics to suggest relevant products tailored to individual preferences of Aadima Store [5]. These gaps in performance, real-time inventory management, and personalization underscore the need for more responsive, intelligent, and user-centric e-commerce solutions. The online Shoe Store aims to address these challenges by offering faster navigation, accurate stock visibility through dynamic database updates, and an enhanced user experience that lays the foundation for future integration of advanced recommendation features.

CHAPTER 3:
SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

This system is designed following the iterative waterfall model of SDLC.

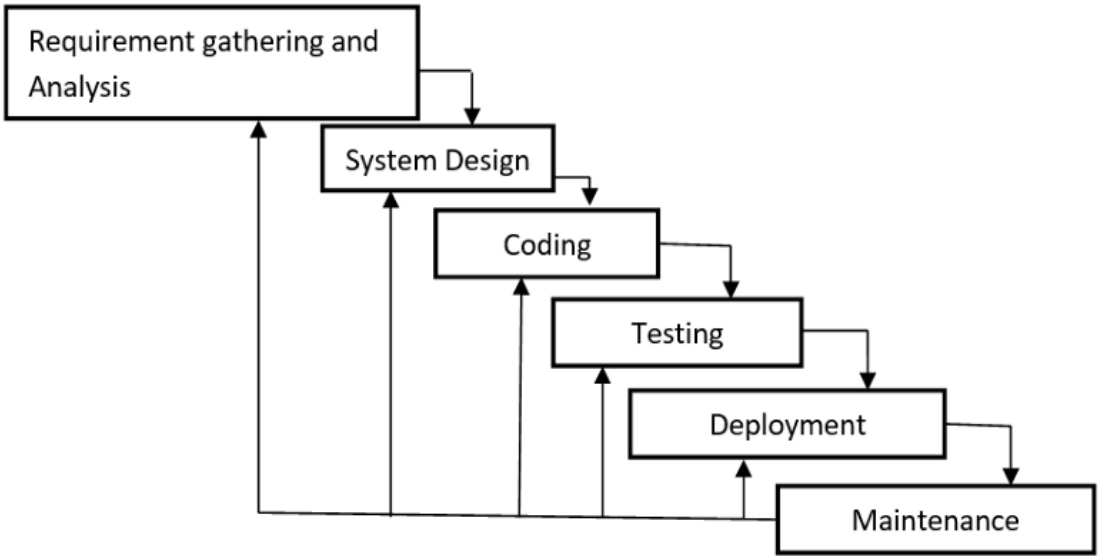

Figure 3-1: Iterative waterfall model

We are using the Iterative Waterfall model because it allow for feedback and revisions at each stage, making it more flexible and easier to change throughout the development process. Here are few key reasons why we use it:

- Can go back and make changes if needed.
- Can catch and fix issues early.
- Test and get feedback regularly to improve.
- Handle big risks early to keep the system safe.

3.1.1 Requirement Analysis

i. Functional Requirements

Functional requirements define the capabilities and function that a system must be able to perform successfully. The functions are very simple and effective as showed in below lines. Users and admin can perform their specific role. The functional requirements of the Shoes Store include:

- **User Registration & Login** – Users should be able to sign up and log in securely.
- **Admin Login** – Admin should log in with secure credentials.
- **Product Browsing & Search** – Users can browse shoes and search by brand.

- **Shopping Cart** – Users can add, remove, or update products in their cart.
- **Checkout & Payment** – Users can place orders and make payments securely.
- **Order Tracking** – Users can view their order history and current order status.
- **Product Management** – Admin can add, edit, or delete shoe details (name, price, stock).
- **User Management**– Admin can view and delete users.
- **Sales & Order Management**– Admin can view all orders, payments and manage refunds or cancellations.
- **Report Generation**– Admin can generate sales and performance reports.

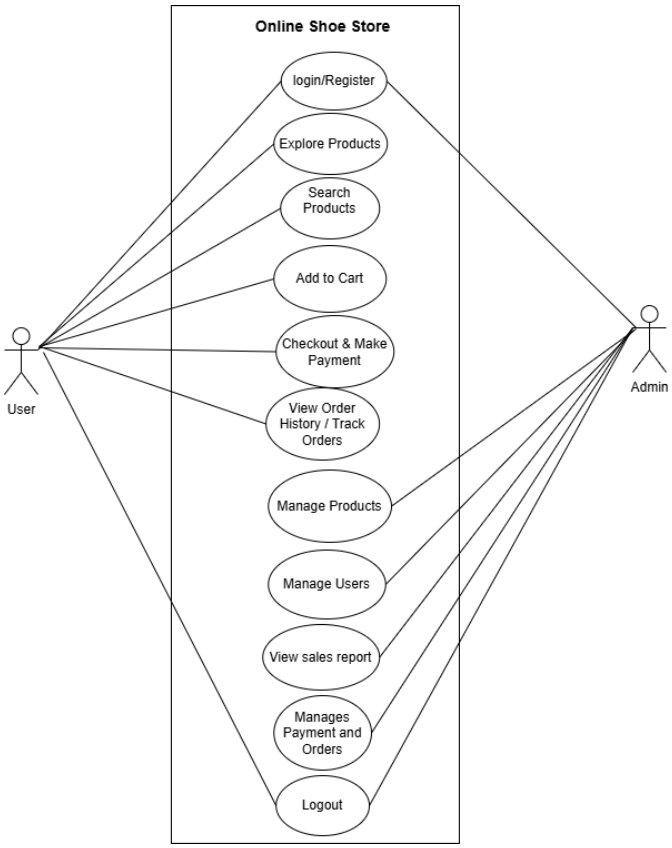

Figure 3-2: Use Case diagram of OSS

ii. Non-Functional Requirements

Non-functional requirements describe how a system must behave and establish formality of its functionality. Our System can perform smoothly and secured Based on our project nonfunctional requirement are described below:

- **Performance** – The system should handle multiple users browsing and purchasing at the same time without lag.
- **Security** – User data and payments should be encrypted, with secure login and transaction methods.

- **Scalability** – The system should be able to handle growth in users, products, and transactions over time.
- **Usability** – The interface should be simple, intuitive, and user-friendly for both customers and admins.
- **Reliability** – The system should function correctly and consistently, minimizing downtime.
- **Maintainability** – The system should be easy to update (e.g., adding new products or features).
- **Portability** – The system should be accessible on multiple devices (desktop, tablet, mobile).

3.1.2 Feasibility Analysis

A feasibility study of online shoes store involves analyzing whether the proposed system is viable from various perspectives, such as technical, operational, economic, legal, and scheduling. The study helps to identify potential risks, costs, and benefits associated with implementing the system

i. Technical Feasibility

While developing this online shoe store, it is needed to consider technical aspect. Some of the technical tools and technologies are listed below.

- PHP
- MySQL
- HTML
- CSS
- JavaScript
- Diagram Drawing Tools(Draw.io)

Most of the tools and technologies mentioned above are freely available. Initially this web application runs on the local host. So it is clear that this online shoe store technically feasible.

ii. Operational Feasibility

After the successful completion of technical feasibility, the operational feasibility analysis is conducted. The system should be easy for user/customers to use without much training. The design will be simple and user-friendly. It will also be accessible to everyone, including people with limited knowledge. The new system should fit well into the current shoes store without causing disruptions.

iii. Economic Feasibility

The system's development come at a zero cost because the project is smaller in scope. Additionally the system is developed by two students and is developed locally so there is no need for spending money on domain and hosting and even hiring developers as well. All the tools and technologies required to develop this system are also freely available. Thus, the Online shoe store project is economically feasible..

iv. Schedule Feasibility

The system is also very much feasible in term of schedule. The particular time period is divided for each phase and we will complete the project within the given time period. Every phase like requirement gathering and analysis, system design, coding, testing, deployment and documentation has their particular time period and have each documentation period.

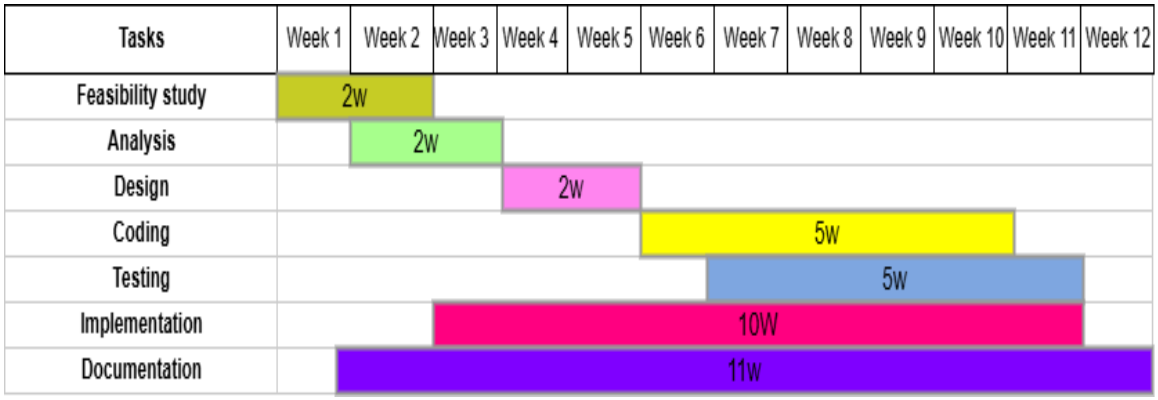

Figure 3-3: Gantt chart of OSS

Feasibility Study(Week 1-2)

The project will starts with a two-week feasibility study to analyze whether the Shoe Store system is practical in terms of technology, cost, and time. This phase ensures that the project goals are realistic and achievable.

System Analysis (Week 2–3)

Following the feasibility study, a two-week analysis is carried out. In this phase, system requirements are gathered, business processes are studied, and both user and admin needs are identified. The analysis helps form a clear picture of what the system should deliver.

System Design(Week 4-5)

Once the requirements are defined, the design phase begins. This two-week phase focuses on creating the database design, system architecture, and user interface design. It sets the blueprint for development.

Coding / Development (Week 6–10)

The coding phase spans five weeks, where the actual system is developed. We implement both the user and admin modules, including product browsing, search, checkout, payment, and admin management features.

Testing (Week 7-11)

Testing overlaps with coding to ensure that each module is checked as soon as it is developed. This five-week phase identifies and resolves bugs, ensuring that the Shoe Store application works correctly, securely, and efficiently.

Implementation (Week 3–12)

Implementation runs throughout the project once the initial system components are ready. It lasts ten weeks and involves integrating modules, deploying them to the working environment, and preparing for real-time usage by users and admins.

Documentation (Week 2–12)

Documentation is an ongoing process lasting eleven weeks. It includes recording project progress, system designs, user manuals, and technical guides. Proper documentation ensures that the project is well-documented for future maintenance and user training.

3.1.3 Data Modelling(ER-Diagram)

An Entity-Relationship (E-R) diagram visually shows the entities, their attributes, and the relationships between them in a database. It helps in designing and understanding the structure of a database. Below is the E-R Diagram for the Online shoe store (OSS).

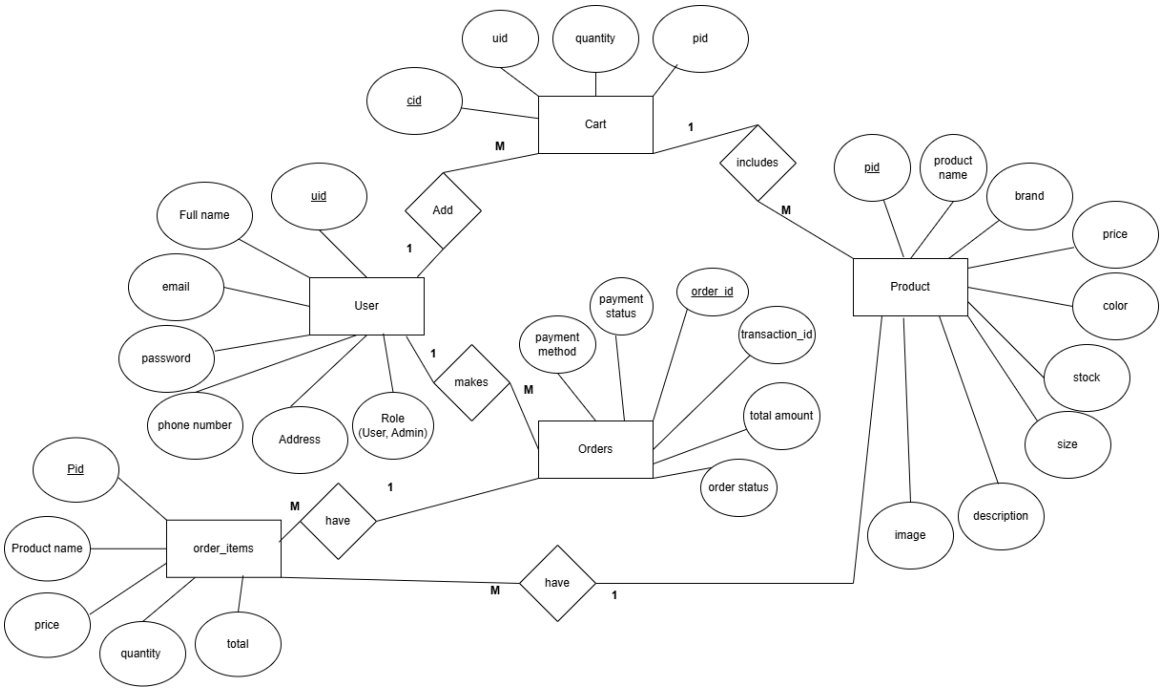

Figure 3-4: Entity Relationship Diagram of OSS

3.1.4 Process Modelling (DFD)

Data Flow Diagram (DFD) of Online shoe store are drawn below:

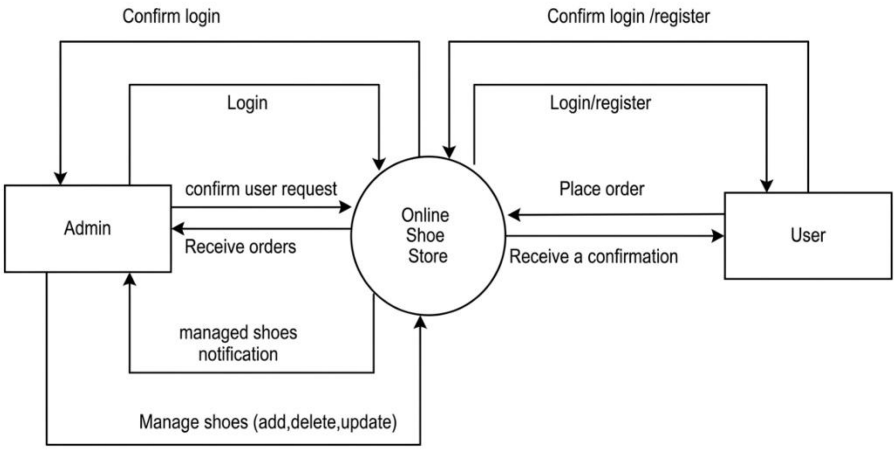

Figure 3-5: Level 0 DFD of OSS

The Level 0 Data Flow Diagram represents the entire Online Shoe Store system as a single process and shows its interaction with external entities. In this diagram, the system interacts mainly with two entities: the User and the Admin. The user performs actions such as login or registration, browsing products, and placing orders, while the admin manages the system by handling products and monitoring orders. The data flows include login confirmation, order placement, order confirmation, and product management information. This level does not show internal processes but provides a high-level overview of how the system communicates with external users and administrators.

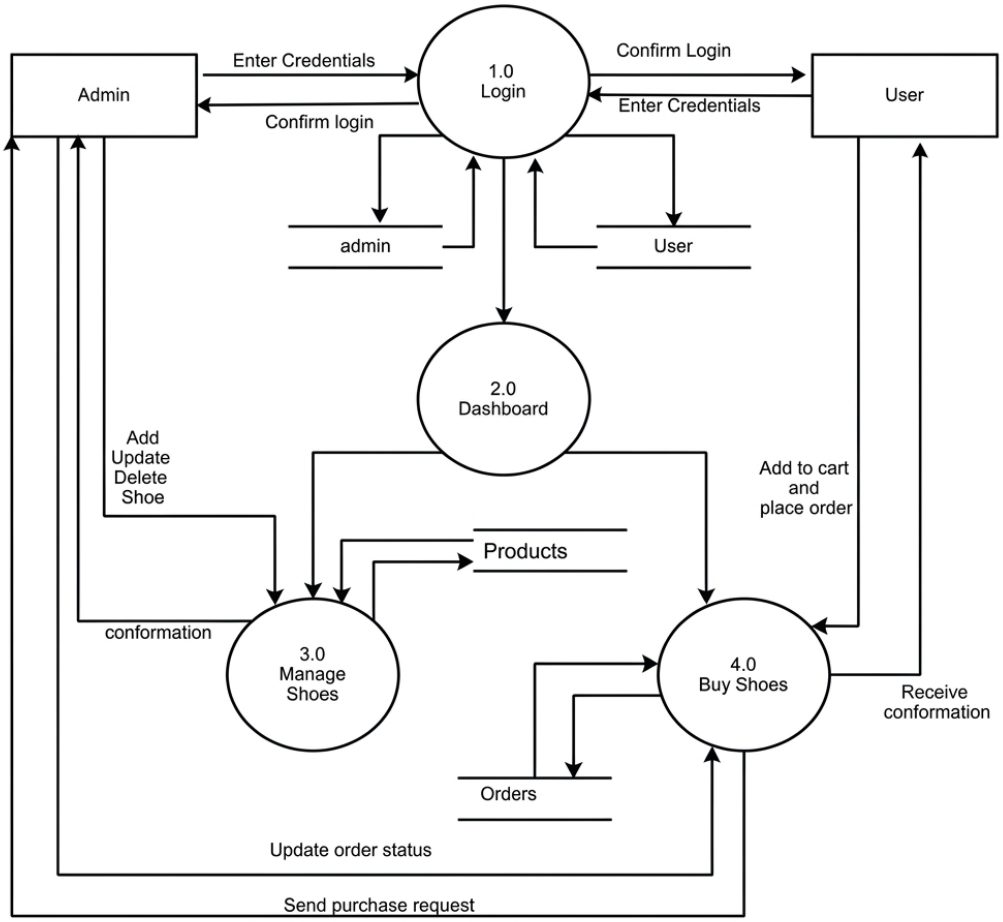

Figure 3-6: Level 1 DFD of OSS

The Level 1 Data Flow Diagram breaks down the entire system into its main functional processes, providing a more detailed view compared to Level 0. The system is divided into four primary processes: Login (1.0), Dashboard (2.0), Manage Shoes (3.0), and Buy Shoes (4.0). The Login process handles authentication of both users and admins. The Dashboard acts as a central interface where users and admins can access different functionalities. The Manage Shoes process allows the admin to add, update, or delete product details, while the Buy Shoes process enables users to browse products, add items to the cart, and place orders. This level also includes data stores such as Products and Orders, showing how data is stored and retrieved within the system. Overall, Level 1 illustrates the main components and data flow between processes, users, and data storage.

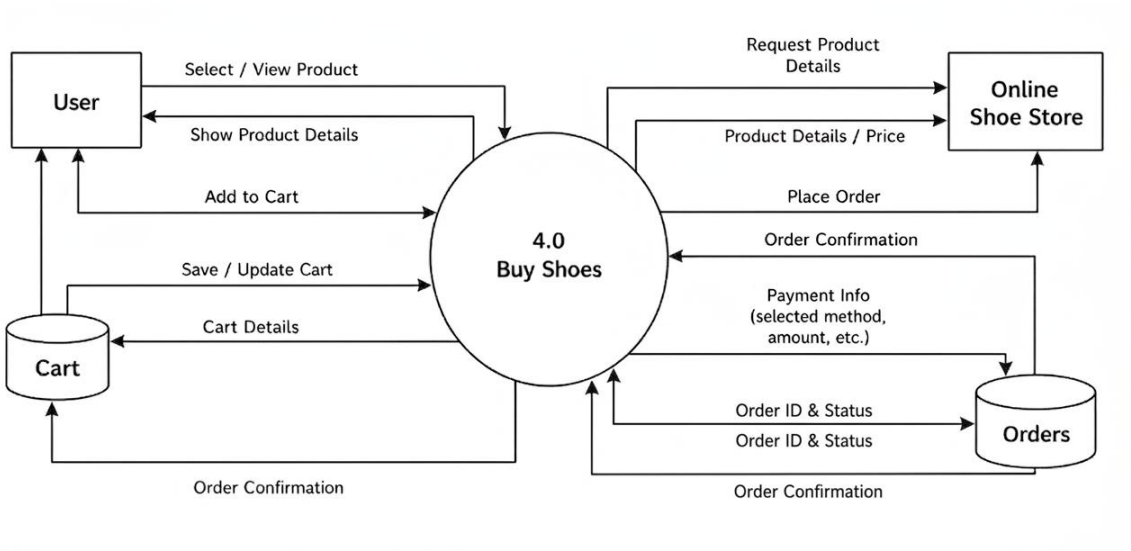

Figure 3-7: Level 2 DFD of OSS

The Level 2 Data Flow Diagram provides a detailed breakdown of the Buy Shoes (4.0) process. It explains how a user interacts with the system step-by-step to complete a purchase. The process begins with viewing product details, followed by adding items to the cart and updating the cart if needed. The user then proceeds to checkout, selects a payment method, and places the order. Finally, the system sends an order confirmation to the user. During this process, the cart temporarily stores selected items, and all order and payment-related information is stored in the Orders database. This level clearly shows the internal workflow and detailed data movement involved in completing a purchase in the system.

3.2 System Design

To realize the different functional requirement of the system in graphical form, different design diagram of the system has been prepared which are as follows:

3.2.1 Architectural Design

For this system, three tier architecture is used which includes user interface (HTML, CSS, JS) , web server(PHP) and database(MySQL). In architectural design, basic structure of the system is shown.

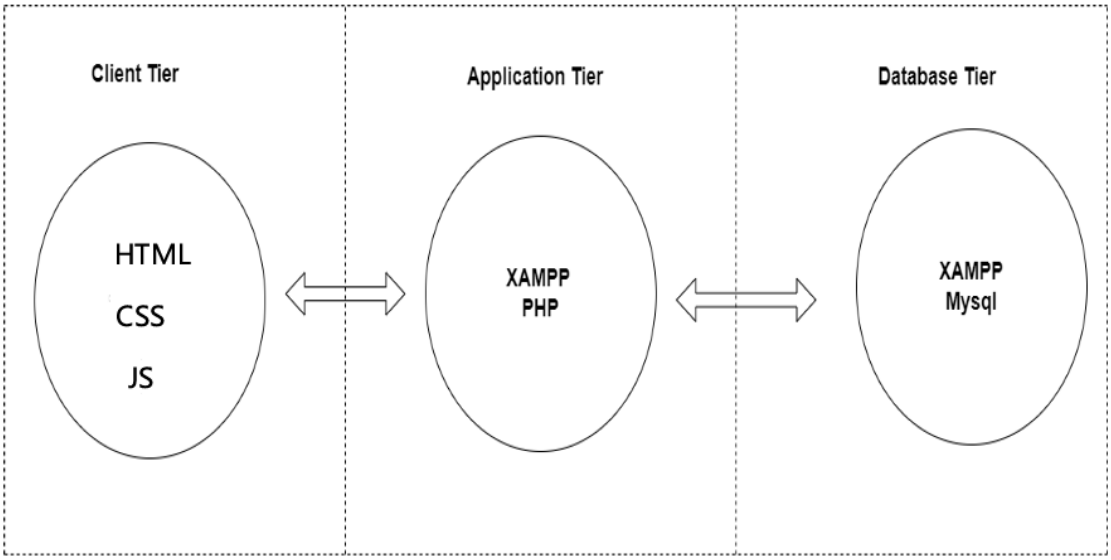

Figure 3-8: Architectural design of OSS

3.2.2 Database Schema Design

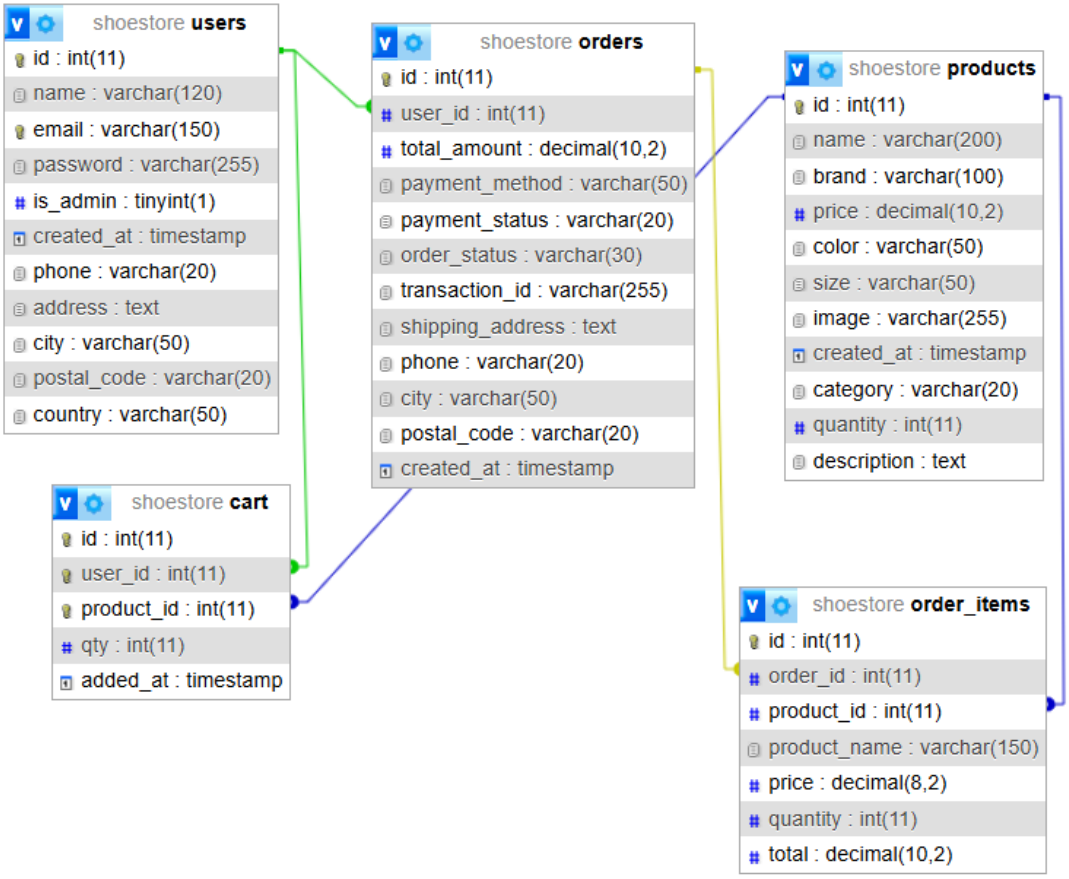

Figure 3-9: Database Schema Design for OSS

3.2.3 Interface Design

The UI/Interface is one of the most important parts of the system as it determines how easy it is for a new user using the system to understand the different components listed and navigate through them in order to achieve the intended goal of using the system. The interface of this system are represented at the appendices section below.

3.2.4 System Flowchart

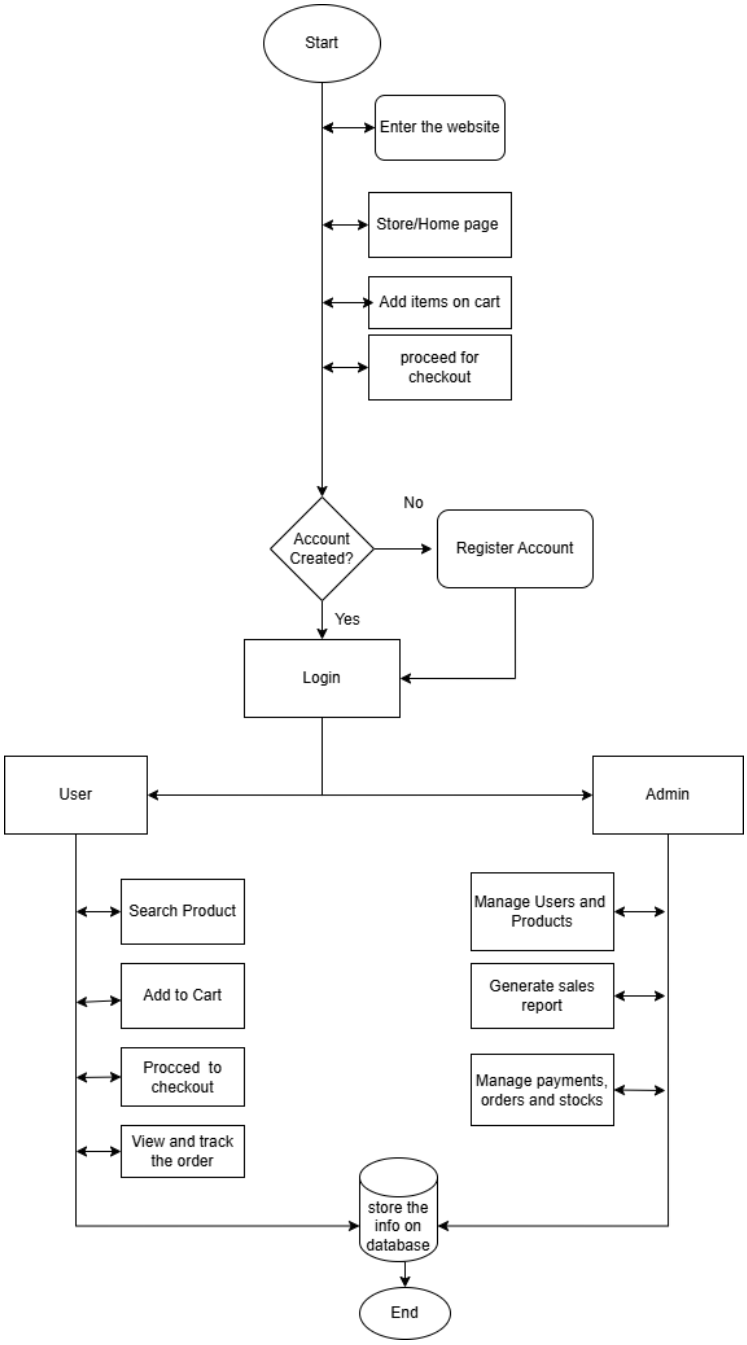

Figure 3-10: Flowchart of Online shoe store

3.3 Algorithm Implementation

The algorithm that are used on my project are:

1. **Password Hashing**

A custom password hashing algorithm is implemented to securely store user passwords in the system. It uses salt-based hashing with SHA-256 and multiple iterations to enhance security and protect against brute-force and rainbow table attacks.

Steps:

Step 1: Generate a random salt using **random_bytes()** to ensure uniqueness of each password.

Step 2: Combine the generated salt with the user's plain password to form a single string.

Step 3: Apply SHA-256 hashing on the combined salt and password to generate an initial hash value.

Step 4: Repeat the hashing process 1000 times by continuously hashing the previous hash with the combined string to increase computational complexity.

Step 5: Extract the final hash and trim it to a fixed length of 28 characters.

Step 6: Store the final result by concatenating the salt and hashed value (salt + hash) in the database for future verification.

Step 7: During login, extract the salt from the stored value and repeat the same hashing process on the entered password.

Step 8: Compare the newly generated hash with the stored hash to validate the user's password.

Code:

```
<?php
function generateSalt($length = 16) {
    return bin2hex(random_bytes($length));
}
function customHashPassword($password, $salt = null) {
    if (!$salt) $salt = generateSalt(16);
    $combined = $salt . $password;
    $hash = hash('sha256', $combined);
    for ($i = 0; $i < 1000; $i++) {
        $hash = hash('sha256', $hash . $combined);
    }
```

```
    $finalHash = substr($hash, 0, 28);
    return $salt . $finalHash;
}
function verifyCustomPassword($password, $stored) {
    $salt = substr($stored, 0, 32);
    return customHashPassword($password, $salt) === $stored;
}
?>
```

2. **Brand Searching(Binary search)**

A modified binary search algorithm is implemented to efficiently find products based on brand names. It is used to quickly locate products whose brand matches a given prefix in a sorted list, improving search performance in the system.

Steps:

Step 1: Initialize the search boundaries (low and high) based on the size of the sorted product array.

Step 2: Convert the search keyword (brand name) to lowercase to ensure case-insensitive comparison.

Step 3: Calculate the middle index (mid) of the current search range.

Step 4: Compare the brand at the middle index with the search keyword:

- If the brand starts with the given prefix, store the index as a potential match and continue searching in the left half to find an earlier occurrence.
- If the middle brand is lexicographically smaller than the search keyword, move the low pointer to the right half.
- Otherwise, move the high pointer to the left half.

Step 5: Repeat the process until the search range is exhausted.

Step 6: Return the index of the first matching brand occurrence if found; otherwise, return -1.

Code:

```
<?php
function binarySearch(array $arr, string $elem): int {
    $low = 0;
    $high = count($arr) - 1;
    $elem = strtolower($elem);
```

```
    $found_index = -1;
    while ($low <= $high) {
        $mid = intdiv($low + $high, 2);
        $midVal = strtolower($arr[$mid]['brand']);
        if (strpos($midVal, $elem) === 0) {
            $found_index = $mid;
            $high = $mid - 1;
        } elseif ($midVal < $elem) {
            $low = $mid + 1;
        } else {
            $high = $mid - 1;
        }
    }
    return $found_index;
}
?>
```

1. **Recommendation Algorithm (Content-Based Filtering)**

A content-based filtering algorithm is implemented to recommend products based on similarity in product category. This method helps in suggesting relevant products to users by analyzing product attributes stored in the database.

Steps:

Step 1: Connect to the database and retrieve all product records from the products table.

Step 2: Store all retrieved products into an array for processing.

Step 3: Filter the products by comparing the category of each product with the selected product's category.

Step 4: Exclude the currently viewed product from the recommendation list to avoid repetition.

Step 5: Reindex the filtered array to maintain proper sequence after filtering.

Step 6: Shuffle the filtered products to add randomness and improve variety in recommendations.

Step 7: Return a limited number of products (e.g., 4) as final recommendations.

Code:

```
function get_category_recommendations($category, $exclude_id = null, $count = 4) {
```

```
global $conn;
$all_products = [];
$result = $conn->query("SELECT * FROM products");
if($result && $result->num_rows > 0) {
    while($row = $result->fetch_assoc()) {
        $all_products[] = $row;
    }
}
// Filter by category and exclude current product
$filtered = array_filter($all_products, function($product) use ($category, $exclude_id) {
    if ($exclude_id !== null && $product['id'] == $exclude_id) return false;
    return isset($product['category']) && $product['category'] === $category;
});
```

CHAPTER 4:
IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

Following are the tools and framework used for the accomplishment of this project:

Front end

The following tools were used for frontend:

HTML

In our system, html is used for creating different webpage and sites. It is used to create and structure sections, headings, links, paragraphs using various tags and elements. We also define headers, paragraphs, links, and images by using html.

CSS

CSS is used for designing different tags of html. It is also used to design different component by the help of class and id. Different CSS are used such as inline css, internal css, and external css to design this system. It is used for defining the styles for web pages. By using css, we can control the text color, font style, the spacing between paragraphs, sizing of columns, layout designs, and many more.

JavaScript

It is used for client-side validation and to make dynamic, interactive and responsive web pages. It is used to add dynamic behavior to the webpage and add special effects to the webpage.

Backend

The following tools were used for backend:

PHP

PHP is a server-side scripting language that is embedded in HTML. It is used to manage dynamic content, databases, session tracking, even build entire e-commerce sites. Although it is especially suited to web development, you can also build desktop standalone applications as PHP also has a command-line interface.

MySQL

MySQL is mainly used for the purpose of database. It is a place where data is stored. Even when the device is shut down, data remains the same with the help of databases. MySQL is mostly used to make different tables for different roles and store different values.

XAMPP

XAAMP is used for local server and database to fulfill the need of the project and MySQL is used as local server and database.

4.1.2 Implementation Details of Modules

Some of the modules used in making this system are:

User module

The User Module is designed to handle all customer-related activities within the Shoe Store Management System. It allows users to register and log in securely to the system, manage their profiles, and browse available shoes by categories such as men, women, and kids. Users can view detailed product information including price, size, and availability, add products to the cart, and place orders. This module also supports online payment through eSewa and enables users to track their order and payment status, ensuring a smooth and personalized shopping experience..

Admin module

The Admin Module provides full administrative control over the Shoe Store Management System. It enables administrators to manage products, monitor customer orders, update order and payment statuses, and manage registered users. The admin can view order details, including purchased items and payment information, and take necessary actions such as processing or canceling orders. This module ensures secure system management, efficient order handling, and smooth overall operation of the online shoe store.

Cart module

The Cart Module acts as a temporary storage area for products selected by users before checkout. It allows users to add shoes to the cart, update quantities or sizes, remove items, and view the total cost in real time. The cart data is maintained using session management so that items remain available while the user navigates the website. This module simplifies the purchasing process by allowing users to review their selected items before placing an order.

Product module

The Product Module manages all operations related to shoe products in the system. It allows the administrator to add new products, update existing product details, and remove products when necessary. This module maintains information such as product category, size, price, stock quantity, images, and descriptions. It ensures that accurate and up-to-date product information is displayed to users, helping them make informed purchasing decisions.

4.2 Testing

4.2.1 Test Cases for Unit Testing

Table 4-1: Test case for User registration

SN.	Test Case	Test Description	Test inputs	Expected outcome	Result
1	User Registration	Correct name, email & password	Full name: Sagar Bhattarai email:s.2413695@gmail.com Password:1902155	User must Register successfully	PASS
2	User Registration	Correct name, email & password	Full name: Rose Andrewson email:Rose@gmail.com Password: 1902155	User must Register successfully	PASS

Table 4-2: Test case for User Login

Test case ID	Test Case	Test Description	Test Data	Expected Output	Result
1	User Login	Correct Email and password	Email:s.2413695@gmail.com Password:1902155	Login Successfully	Pass
2	User Login	Invalid password	Email:s.2413695@gmail.com Password:oki123	Incorrect password	Pass

Table 4-3Test Case for Admin Login

Test case ID	Test Case	Test Description	Test Data	Expected Output	Result
1	Admin Login	Correct Email and password	Email:admin@gmail.com Password:1902155	Redirect to admin Dashboard	Pass

4.2.2 System Testing

Table 4-4: System Test for User

SN	Task	Task Description	Expected outcome	Result
1	User Registration and Login	If the user id and password match with the database then redirect to user dashboard.	Successfully register/login and redirect Home page.	Redirect to Home page
2	Add to Cart	User explore their desire shoes and add to cart.	Cart section shows the shoes that user add	User can see the cart include the product that they add.
3	Proceed for checkout	Here user can proceed for their payment of the product, Payment are done through Online (esewa) and Cash on delivery.	Shows payment method .	Successfully displayed the payment method with online(esewa) and cash on delivery.
4	Payment through esewa	User proceed the online payment through esewa and give their valid credentials and payed the bill of their product	Open esewa portal and payment is success.	Payment with esewa is successfully done
5	Order Track	User can track their order from order section	Shows the list of orders that the user has ordered.	Display the order lists.

Table 4-5 System test for Admin

SN	Task	Task Description	Expected outcome	Result
1	Admin Login	If the admin email and password match with the database then redirect to admin dashboard.	Successfully login and redirect admin page.	Redirect to admin page.
2	Add shoes	Admin can add the Shoes by filling the Product name, Description, Price, Stock, Category, Brand, Color, Size and product image	New product added successfully	Product added successfully
3	Edit Shoes	Admin can edit any shoes price, brand if necessary to update	Update the shoes details.	Updated successfully.
4	Delete any shoes	If the shoes is no longer available, admin can delete the shoes	Delete the shoes from the product.	Deleted successfully.
5	Manage order	Admin can manage the orders; they can make the status complete, cancelled and pending.	Status updated successfully.	Status updated successfully.

CHAPTER 5:
CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Lesson Learnt

This project was helpful in teaching some important lessons.

- Learned how to build a website for online shopping.
- Gained experience in creating and managing a database.
- Learned to add, remove, and manage products in a shopping cart.
- Understand how online shopping works from browsing to payment

5.2 Conclusion

In conclusion, this project aims to simplify the shopping process by providing a digital platform that reduces the challenges and time required in traditional shoe shopping. It offers a user-friendly interface where customers can browse products, add or remove items from the cart, and place orders conveniently online. The admin panel allows efficient store management, enabling administrators to add, edit, or remove products and manage customer orders easily. This system enhances accessibility, convenience, and efficiency in buying and selling shoes.

5.3 Future Recommendations

Some of the features that can be implemented in the future can be:

- Add discounts, offers, and coupon codes to attract more customers.
- Implement order tracking so customers can track delivery status.
- Introduce wishlist or favorites to save products for later.
- Include customer support chat or helpdesk for assistance.

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[1] F. D. (. Davis, "User Acceptance of E-Commerce Platforms," *Journal of Information Systems,* vol. 34, no. 2, pp. 45-58, 2019.

[2] makebusiness.eu, "The Importance of Secure Payment Gateways in E-commerce," [Online]. Available: https://makebusiness.eu/the-importance-of-secure-payment-gateways-in-e-commerce/. [Accessed 01 dec 2025].

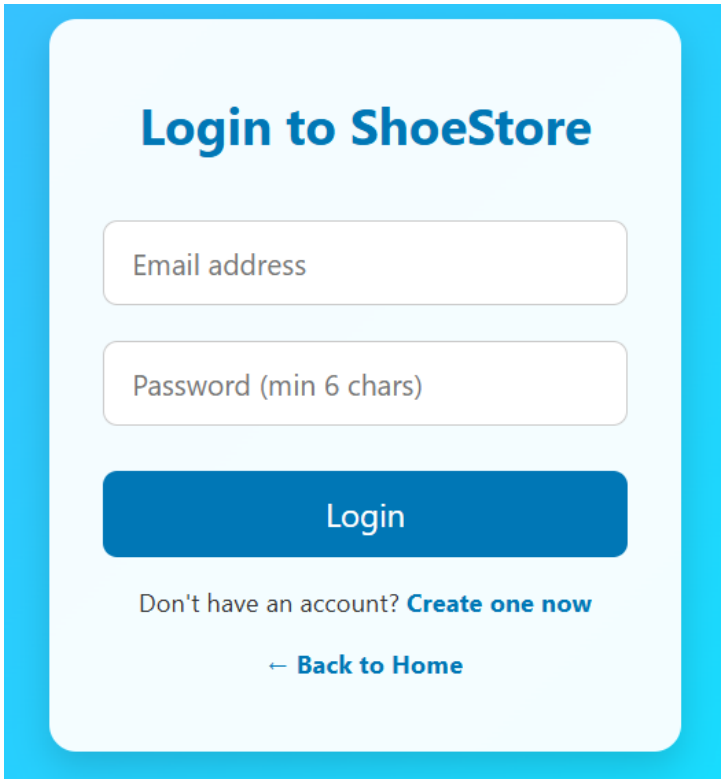
[3] Macys, "Macys.com," [Online]. Available: https://www.macys.com/shop/mens?id=1. [Accessed 02 dec 2025].

[4] Ekjor. [Online]. Available: https://www.ekjor.com/. [Accessed 02 dec 2025].

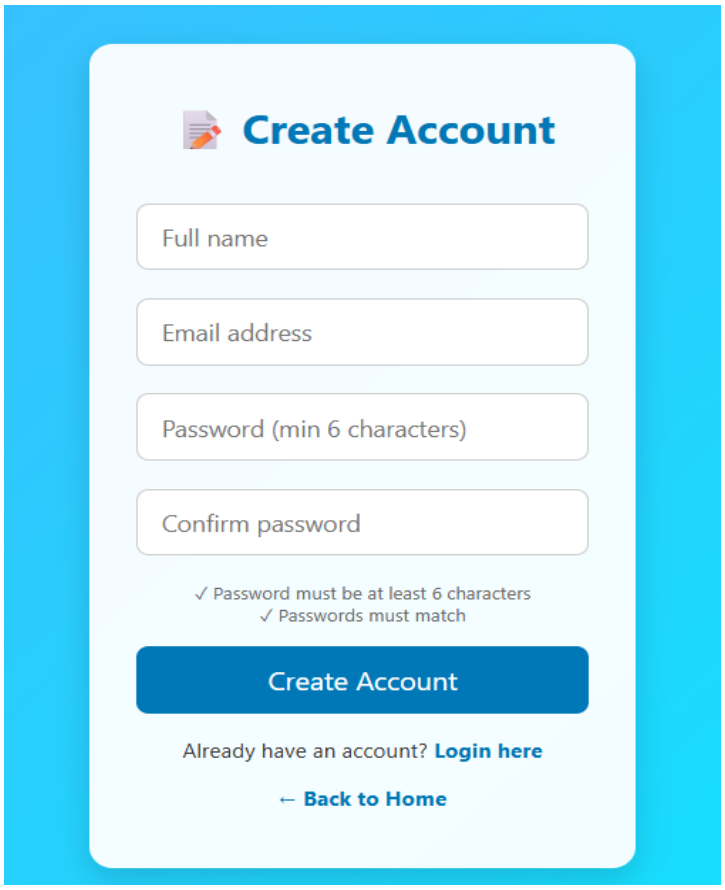
[5] A. Store. [Online]. Available: https://aadima.store/. [Accessed 07 dec 2025].

APPENDICES

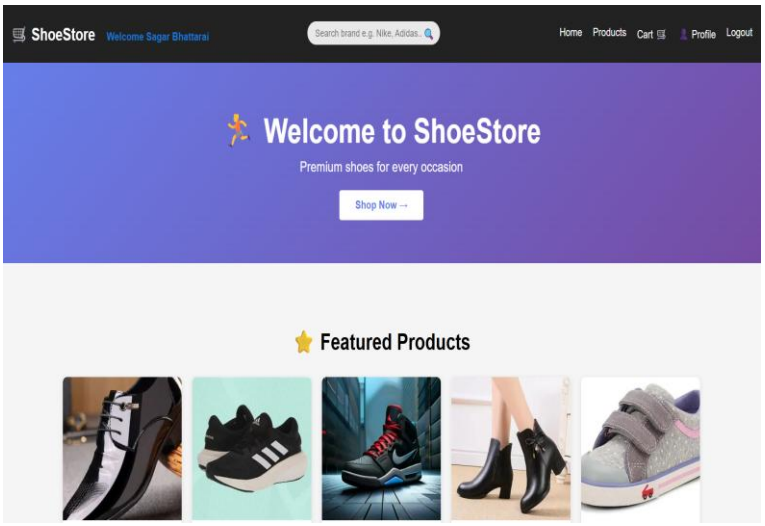
Appendix 1: Login Page



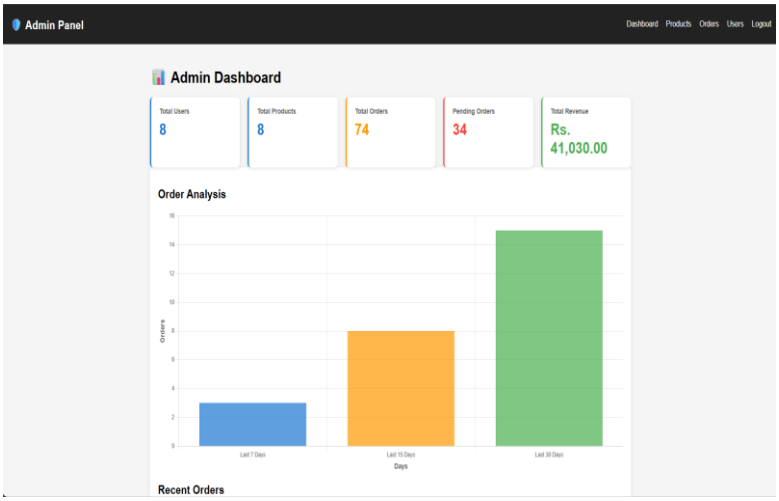
Appendix 2: Register Page



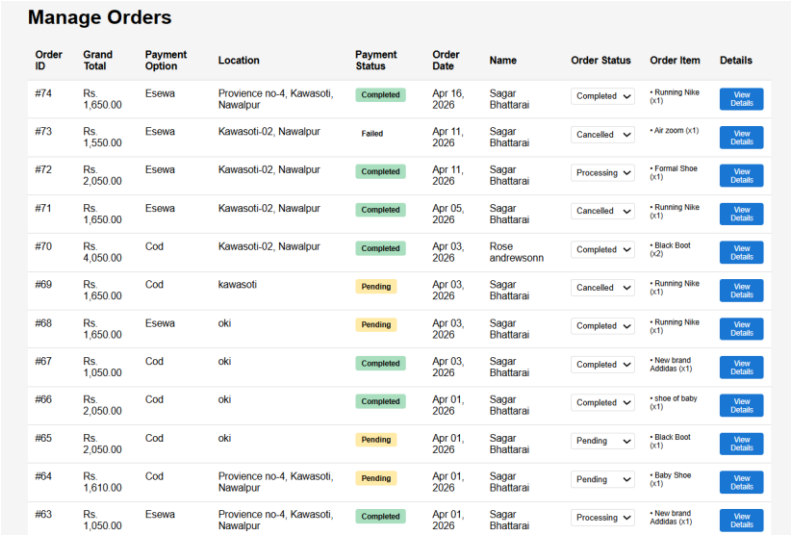
Appendix 3: User Dashboard



Appendix 4: Admin Dashboard



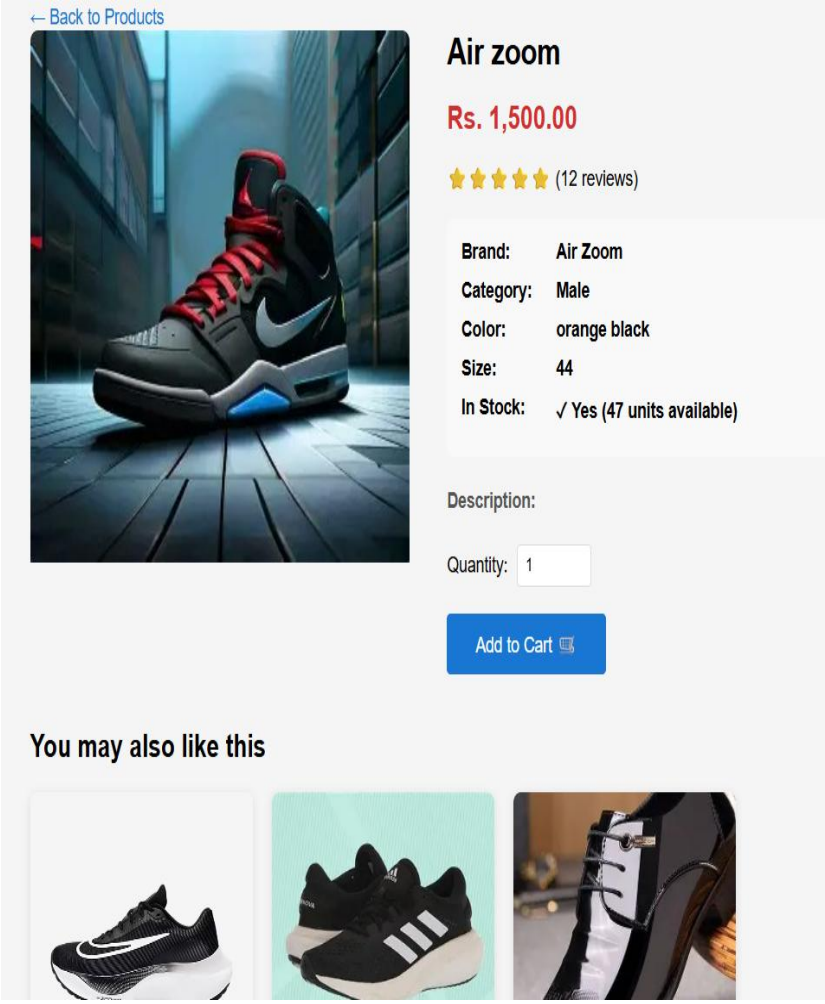
Appendix 5: Manage Orders



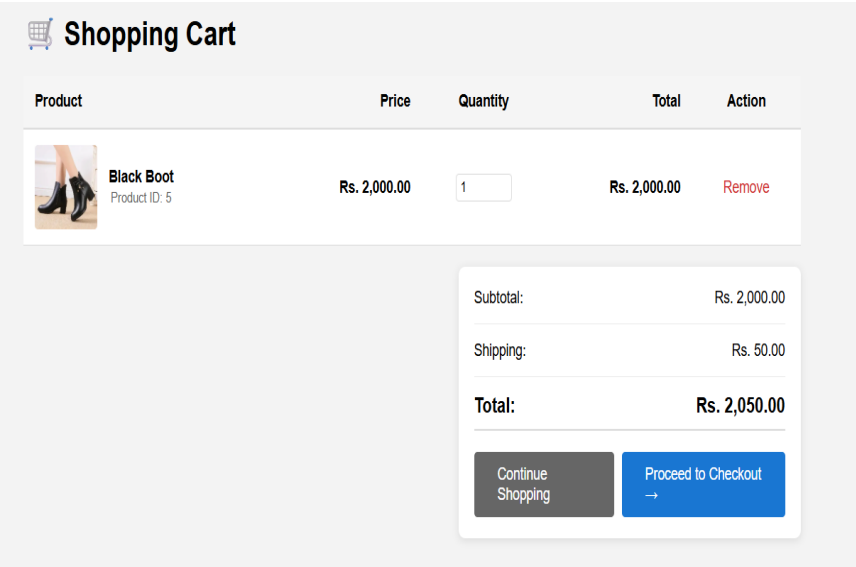
Manage Orders

Order ID	Grand Total	Payment Option	Location	Payment Status	Order Date	Name	Order Status	Order Item	Details
#74	Rs. 1,650.00	Esewa	Provience no-4, Kawasoti, Nawalpur	Completed	Apr 16, 2026	Sagar Bhattarai	Completed	• Running Nike (x1)	View Details
#73	Rs. 1,550.00	Esewa	Kawasoti-02, Nawalpur	Failed	Apr 11, 2026	Sagar Bhattarai	Cancelled	• Air zoom (x1)	View Details
#72	Rs. 2,050.00	Esewa	Kawasoti-02, Nawalpur	Completed	Apr 11, 2026	Sagar Bhattarai	Processing	• Formal Shoe (x1)	View Details
#71	Rs. 1,650.00	Esewa	Kawasoti-02, Nawalpur	Completed	Apr 05, 2026	Sagar Bhattarai	Cancelled	• Running Nike (x1)	View Details
#70	Rs. 4,050.00	Cod	Kawasoti-02, Nawalpur	Completed	Apr 03, 2026	Rose andrewsonn	Completed	• Black Boot (x2)	View Details
#69	Rs. 1,650.00	Cod	kawasoti	Pending	Apr 03, 2026	Sagar Bhattarai	Cancelled	• Running Nike (x1)	View Details
#68	Rs. 1,650.00	Esewa	oki	Pending	Apr 03, 2026	Sagar Bhattarai	Completed	• Running Nike (x1)	View Details
#67	Rs. 1,050.00	Cod	oki	Completed	Apr 03, 2026	Sagar Bhattarai	Completed	• New brand Addidas (x1)	View Details
#66	Rs. 2,050.00	Cod	oki	Completed	Apr 01, 2026	Sagar Bhattarai	Completed	• shoe of baby (x1)	View Details
#65	Rs. 2,050.00	Cod	oki	Pending	Apr 01, 2026	Sagar Bhattarai	Pending	• Black Boot (x1)	View Details
#64	Rs. 1,610.00	Cod	Provience no-4, Kawasoti, Nawalpur	Pending	Apr 01, 2026	Sagar Bhattarai	Pending	• Baby Shoe (x1)	View Details
#63	Rs. 1,050.00	Esewa	Provience no-4, Kawasoti, Nawalpur	Completed	Apr 01, 2026	Sagar Bhattarai	Processing	• New brand Addidas (x1)	View Details

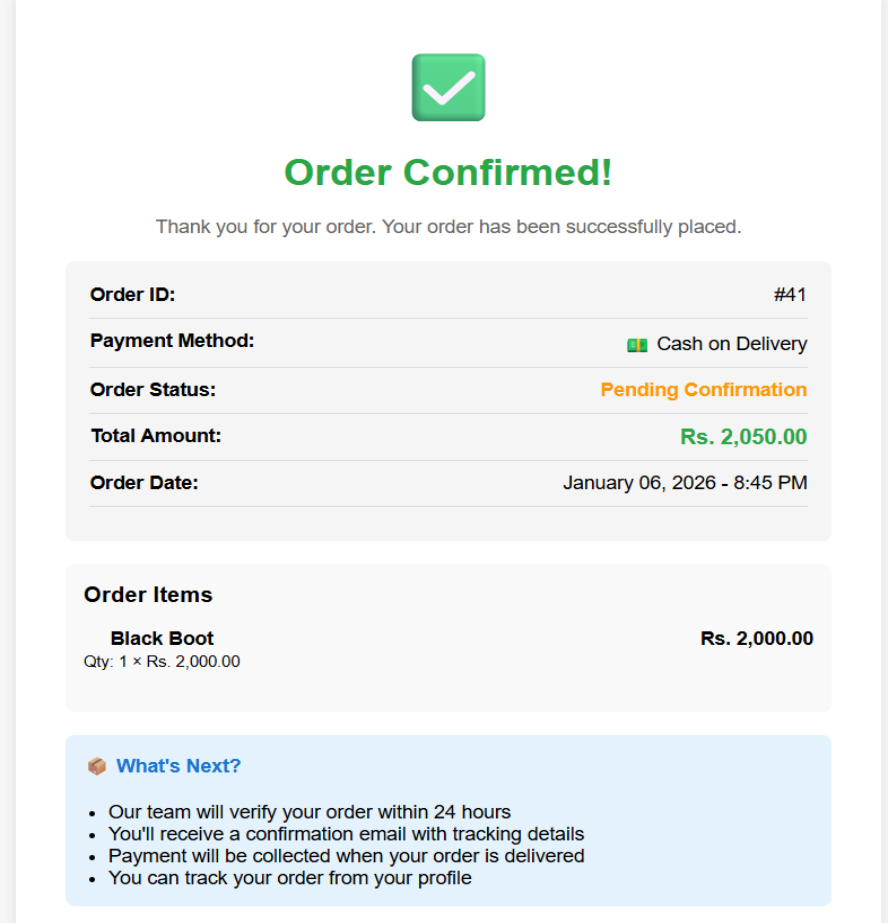
Appendix 6: User viewing product details



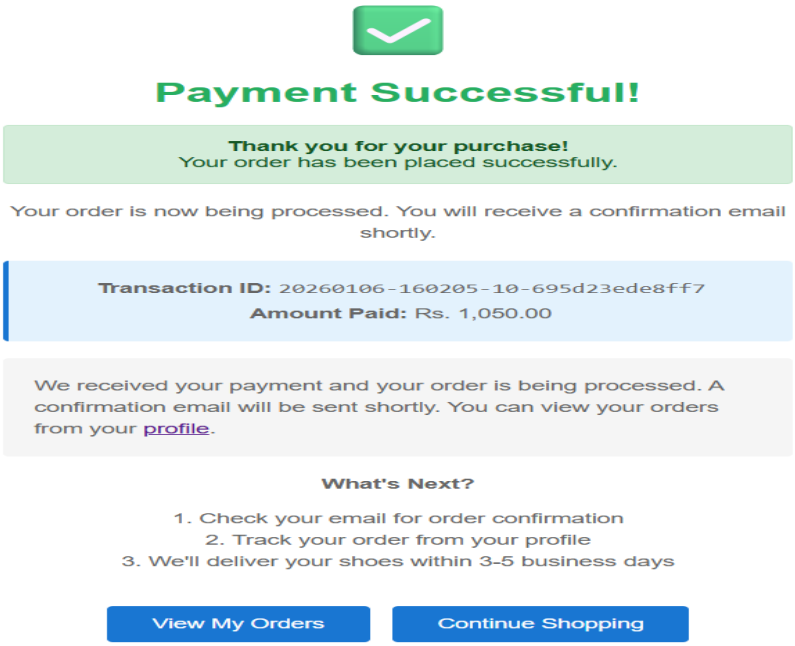
Appendix 7: user proceed to checkout



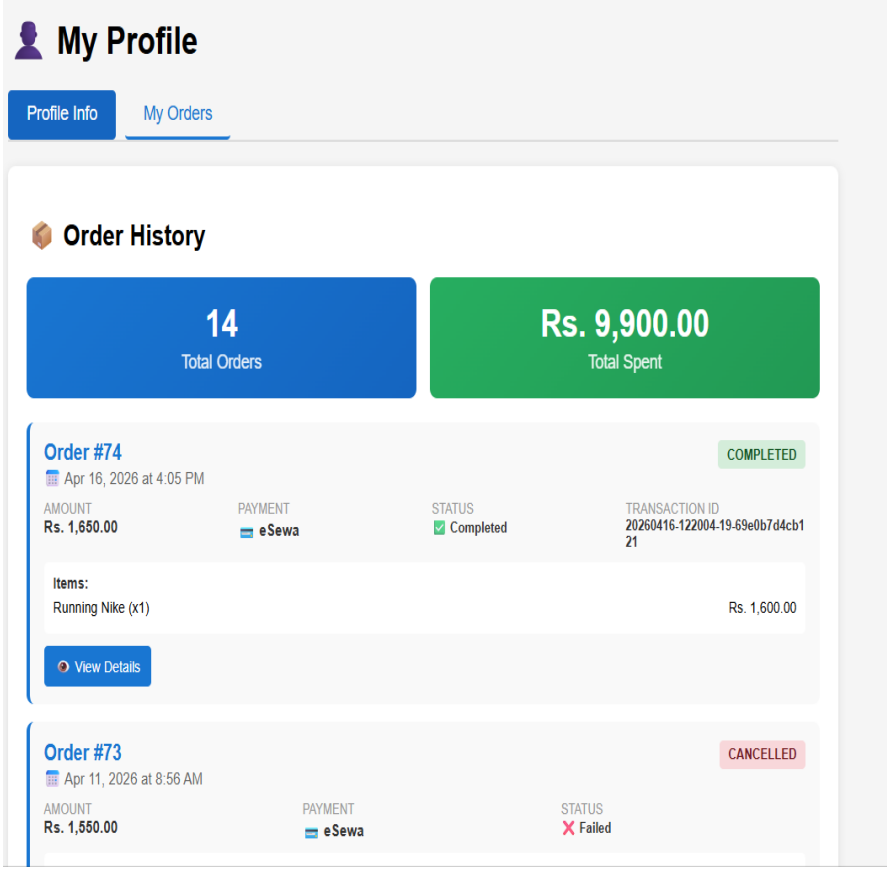
Appendix 8: Cash on delivery



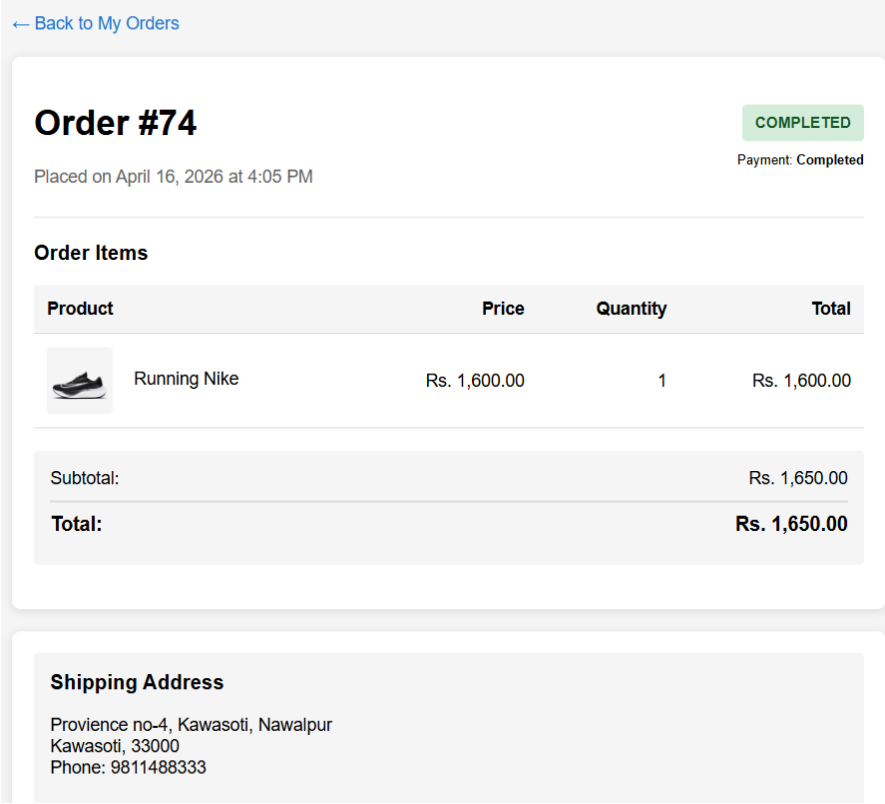
Appendix 9: Payment through Esewa



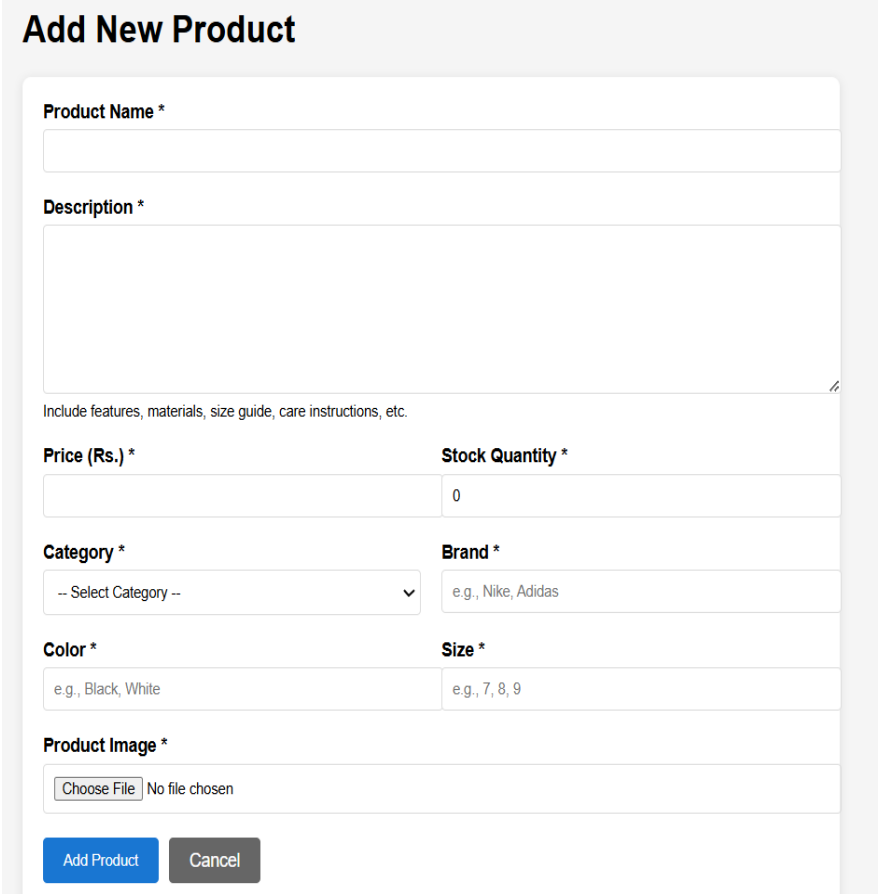
Appendix 10: User order list



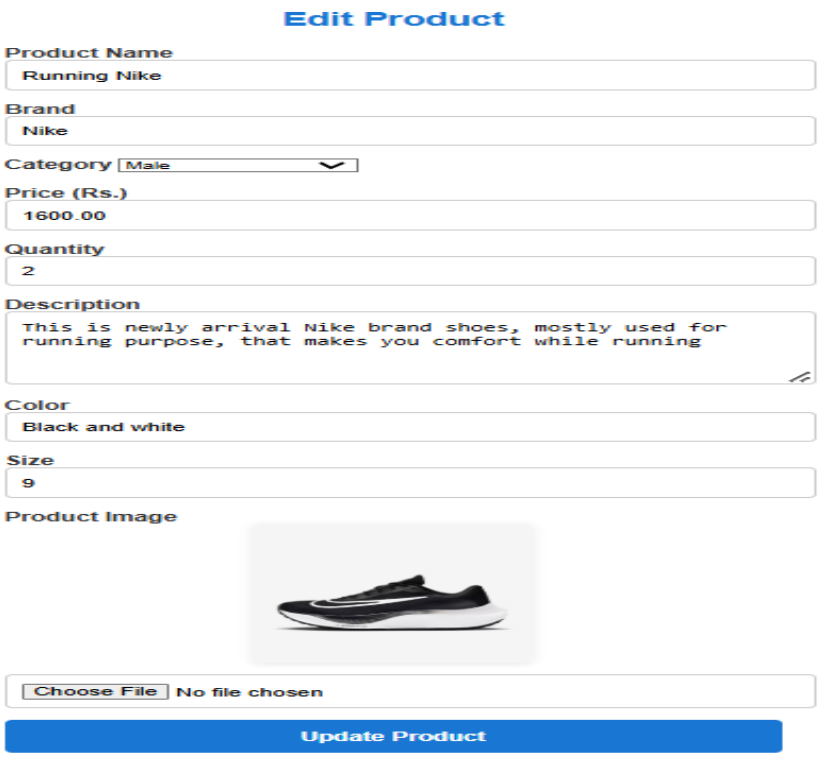
Appendix 11: User's order Details



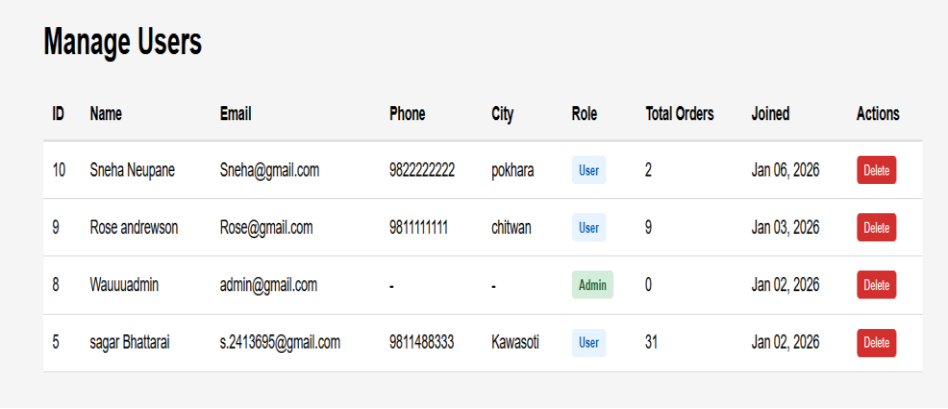
Appendix 12: Admin Add new product



Appendix 13: Admin Edit product



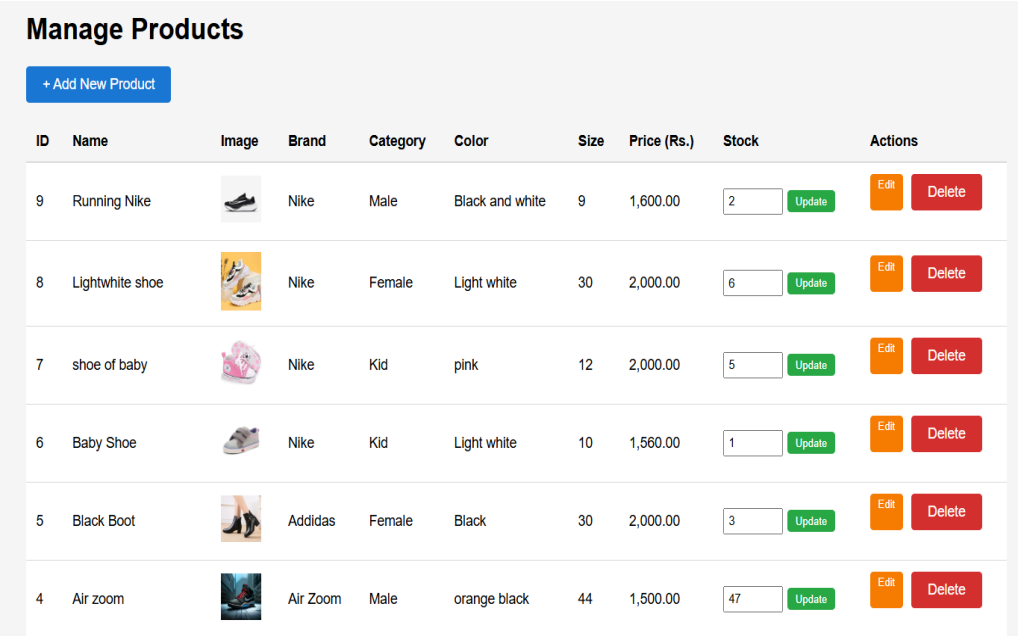
Appendix 14: Admin can delete and saw the details of user



Manage Users

ID	Name	Email	Phone	City	Role	Total Orders	Joined	Actions
10	Sneha Neupane	Sneha@gmail.com	9822222222	pokhara	User	2	Jan 06, 2026	Delete
9	Rose andrewson	Rose@gmail.com	9811111111	chitwan	User	9	Jan 03, 2026	Delete
8	Wauuuadmin	admin@gmail.com	-	-	Admin	0	Jan 02, 2026	Delete
5	sagar Bhattarai	s.2413695@gmail.com	9811488333	Kawasoti	User	31	Jan 02, 2026	Delete

Appendix 15: Admin managing the products and stock



Manage Products

+ Add New Product

ID	Name	Image	Brand	Category	Color	Size	Price (Rs.)	Stock	Actions
9	Running Nike		Nike	Male	Black and white	9	1,600.00	2 Update	Edit Delete
8	Lightwhite shoe		Nike	Female	Light white	30	2,000.00	6 Update	Edit Delete
7	shoe of baby		Nike	Kid	pink	12	2,000.00	5 Update	Edit Delete
6	Baby Shoe		Nike	Kid	Light white	10	1,560.00	1 Update	Edit Delete
5	Black Boot		Addidas	Female	Black	30	2,000.00	3 Update	Edit Delete
4	Air zoom		Air Zoom	Male	orange black	44	1,500.00	47 Update	Edit Delete